TED (21) – 2022 REVISION 2021

FOURTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING AND TECHNOLOGY

PLASTIC TECHNOLOGY

MODEL QUESTION PAPER – SET-1

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	Classify the following polymer with respect to thermal response PC, Epoxy, PF, PET	M1.02	R
2	Mention the principle behind the production of toughened poly styrene?	M2.02	U
3	Write the catalyst or initiator used for the preparation of PS, & isotactic PP?	M2.02, M2.01	R
4	Distinguish thermoplastic polyesters and thermo set polyesters?	M4.03	U
5	Write the names of monomers of poly carbonates? .	M3.02	R
6	Give an example for natural plastic.	M1.02	U
7	Define eco friendly plastics	M1.02	R
8	Write the structure of Teflon	M3.04	R
9	List any two commodity plastics.	M1.02	R

PART B

II. Answer any Eight questions from the following

(8 x 3 = 24 Marks)

		Module Outcome	Cognitive level
1	What are the advantages and disadvantages of plastics over conventional materials?	M 1.01	R
2	Write the procedure for the manufacture of Expanded polystyrene	M2.02	U
3	Write a typical formulation for suspension polymerisation of PVC?	M1.04	U
4	List two important properties and one application of LLDPE	M1.03	R

5	Explain the monomer preparation of HMDA and adipicacid ?	M3.01	U
6	Name the chemicals used for the preparation of epoxy resin. Write the structure of Epoxy resin.	M4.03	U
7	Name the monomers and their structure used for the production UP resin	M4.03	U
8	What are the properties and application of polyurethane	M4.04	U
9	Suggest two applications for PP and give its justification	M2.01	R

PART C

III. Answer all questions from the following (6 x 7 = 42 Marks)

		Module Outcome	Cognitive level
1	Discuss Ziegler process. Explain the properties and applications of HDPE?	M1.03	U
	OR		
2	Explain the manufacture and applications of PVC by suspension polymerisation technique with a formulation?	M1.04	U
3	Explain the manufacture and properties of poly propylene with a flow diagram.	M2.01	U
	OR		
4	Explain the manufacture of PMMA by mass polymerisation technique? Give its properties.	M2.03	U
5	Explain the monomer preparation and polymerization of poly carbonate	M3.02	U
	OR		
6	Write the structure, properties and applications of Teflon.	M3.04	U
7	Describe the raw material, industrial production, and applications of MF resin.	M4.02	U
	OR		
8	Describe industrial production, properties and applications of Novalac resin.	M4.02	U
9	Describe monomer preparation and industrial production of Poly Urethane.	M4.04	U
	OR		
10	Describe the raw material, industrial production, and applications of UF resin	M4.02	U
11	Discuss the role of plastic materials in modern life	M 1.01	U
	OR		

12	Discus the tower process used for the production of PS	M2.02	U

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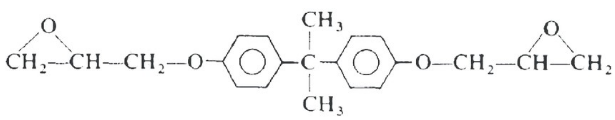
SCHEME OF VALUATION

(Scoring Indicators)

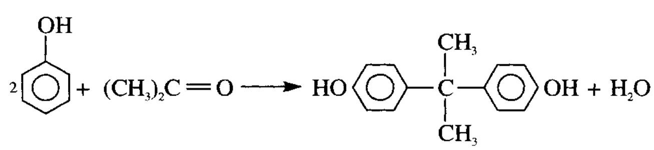
Revision : 2021				**Course Code :**
Course Title : PLASTIC TECHNOLOGY				

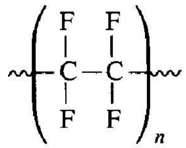
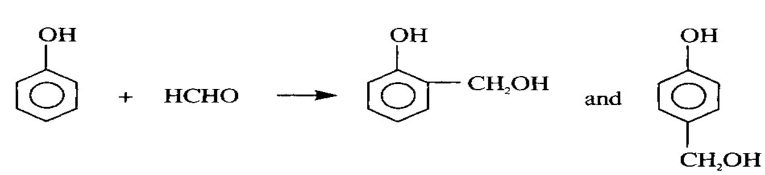
Qst. No	**Scoring Indicator**	**Split up Score**	**Sub Total**	**Total**
I				
1	Thermoplastic-PC,PET.	0.5		
	Thermoset-Epoxy,PF	0.5	0.5*2	1
2	Co polymerisation with another monomer	0.5		
	Blending with amorphous polymer	0.5	0.5*2	1
3	PS-peroxide initiator	0.5		
	Isotactic PP-Z/N catalyst	0.5	0.5*2	1
4	thermoplastic polyesters- saturated polyester	0.5		
	thermo set polyesters- unsaturated polyester	0.5	0.5*2	1
5	Bisphenol-A &Phosgene	0.5		
		0.5	0.5*2	1
6	Cellulose, polypeptides	1	1	1
7	Plastic that are not harmful to environment	1	1	1
8	$-\left[CF_2-CF_2\right]_n-$	1	1	1
9	PP,PE,PVC,PS	1	0.5*2	1
II	advantages			
1	Elelectrical and thermal insulation properties	1.5		

	Water resistanLight in weight • Easily moulded& excellent finishing. • Very good strength and toughness. • Corrosion resistant & chemically inert • Low thermal expansion coefficient • Good electrical t &adhesive nature Low cost &Reused ,recycled (Thermoplastics Disadvantges The raw materials for plastics are non renewable resources. *Plastic burning causes cancerous materials *Deformation under load *Embrittlement at low temperature * Low heat resistant and poor ductility	1.5 1.5	1.5+1.5	3 3
2	The blowing agent, typically 6% of low boiling petroleum ether fraction such as n-pentane, may be incorporated before polymerisation or used to impregnate the bead under heat and pressure in a post-polymerisation operation. The impregnated beads may then be processed by two basically different techniques: (a) the steam moulding process, the most important industrially and (b) direct injection moulding or extrusion .	1.5	1.5+1.5	3
3	A typical charge would be: Monomer --Vinyl chloride 30-50 parts Dispersing agent -Gelatine 0.001 parts Modifier -Trichlorethylene 0.1 parts Initiator -Caproyl peroxide 0.001 parts Demineralised water 90 parts	3	3	3
4	Properties -Good toughness, Good pierce resistance, good chemical stability Application – overhead tanks	1*2 1	2+1	3
5	benzene is the starting material..Benzene on liquid phase hydrogenation at 210°C under pressure produces cyclohexane .It is then oxidised in the presence of Naphthalene catalyst .The product is a mixture of Cyclohexanol (70%)&Cyclohexanone(30%) The mixture is further oxidised with 50%HNO3 at75°C in the presence of copper ammonium vanadate catalyst . Adipic acid has a melting point 151°C .T Adipic acid .A typical route is that via cyclohexane, and cyclohexanol. To produce cyclohexane, benzene is subjected to continuous liquid phase hydrogenation at 3401bf/in2 pressure and a temperature of 210°C	1.5 1.5	3	3

6	using a Raney nickel catalyst. After cooling and separation of the catalyst the produce is fed to the cyclohexane store. In the next stage of the operation for cyclohexane is preheated and continuously oxidised in the liquid phase by air using a trace of cobalt naphthenate as catalyst. This gives an approximately 70% yield of a mixture of cyclohexanol and cyclohexanone with a small quantity of adipic acid. The cyclohexanol-cyclohexanone mixture is converted into adipic acid by continuous oxidation with 50% HN03 at about 75°C using a copper-ammonium vanadate catalyst. The adipic acid is carefully purified by subjection to such processes as steam distillation and crystallisation. The pure material has a melting point of 151°C.	3 1.5 1.5	3 1.5*2	3 3
7	Bisphenol A, Epichlorohydrin 	1.5 1.5	1.5*2	
8	Propylene glycol 146 parts Maleic anhydride 114 parts Phthalic anhydride 86 parts The molar ratios of these three ingredients in the order above is 1.1 : 0.67 : 0.33. Write the structures also. Applications of PU 1.Construction & Building Projects: 2.Transportation & Marine: 3.Furniture and Bedding: 4.Machinery and Foundry: 5.Appliance	1	1+2	3 3
9	Low-molecular-weight polyols or chain-extending diols, will tend to impart rigidity into a PU. It follows that the ratio of polyol to isocyanate is high for high-molecular-weight polyols and low for low-molecular-weight polyols . A low isocyanate index may tend to encourage plasticization of the PU because of the presence of excess polyol, and a high isocyanate index will promote stiffening through cross-linking side reactions . Cellular PUs are relatively cheap to manufacture, and dominate the marketplace, even although the cost of raw materials may be at least twice that for foamed commodity thermoplastics Applications – syringe, chair Syringe – thermal stability above 100 C Chair – Good mechanical properties	3 0.5*2 1*2	3 1+2	3

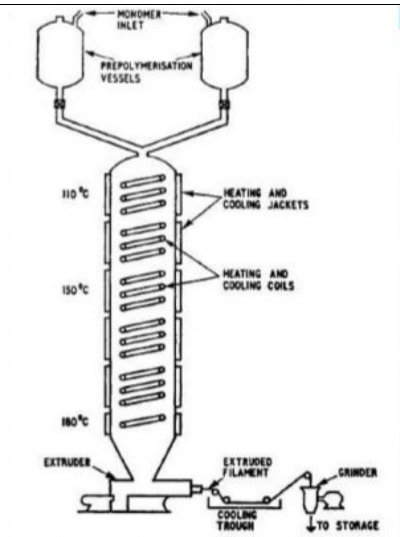
III				
1	In a typical process the catalyst is prepared from titanium tetrachloride and aluminiumtriethyl or some related material. Ethylene is fed under low pressure into the reactor which contains liquid hydrocarbon to act as diluent. The catalyst complex may be first prepared and fed into the vessel or may be prepared *in situ* by feeding the components directly into the main reactor. Reaction is carried out at some temperatures below 100°C (typically 70°C) in the absence of oxygen and water, both of which reduce the effectiveness of the catalyst	7	7	7
2	**A** typical charge would be: Monomer Vinyl chloride 30-50 parts Dispersing agent Gelatine 0.001 parts Modifier Trichlorethylene 0.1 parts Initiator Caproyl peroxide 0.001 parts Demineralised water 90 partsUsing the above recipe the dispersing agent is first dissolved in a knownweight of water, and added to the kettle. The rest of the water, the peroxide andthe modifier are then added to the kettle which is sealed down and evacuated to28 inHg. The vinyl chloride is then drawn in from the weighing vessel. In some casespressurised oxygen-free nitrogen may be used to force all the monomer intothe vessel. This is then closed and heated to about 50°C. Through the rise intemperature a pressure of about 100 lbf/in2 (0.7 MPa) will be developed in thereactor. As reaction proceeds the temperature is maintained until the pressurestarts to fall, due to consumption of the monomer. When the pressure has dropped to 10-20 lbf/in2 (0.07-0.13 MPa) excess monomer is removed, either in theautoclave, for example by a steam stripping process, or outside the autoclave ina column in which the slurry of polymer particles in water passes through acounter current of steam.	2 5	2+5	7
3	There are many points of resemblance between the production of polypropyleneand polyethylene using Ziegler-type catalysts. In both cases the monomers areproduced by the cracking of petroleum products such as natural gas or light oils.For the preparation of polypropylene the C3 fraction (propylene and propane) isthe basic intermediate and this may be separated from the other gases withoutundue difficulty by fractional distillation. The separation of propylene frompropane is rather more difficult and involves careful attention to the design of the distillation plant. For polymer preparation impurities such as water and methylacetylene must be carefully removed. A typical catalyst system may beprepared by reacting titanium trichloride with aluminiumtriethyl, aluminiumtributyl or aluminium diethyl monochloride in naphtha under nitrogen to form aslurry consisting of about 10% catalyst and 90% naphtha. The properties of thepolymer are strongly dependent on the catalyst composition and its particle shape and sizeIn the suspension process, propylene is charged into the polymerisation vessel under pressure whilst the catalyst solution and the reaction diluent (usually naphtha) are metered in separately. In batch processes reaction is carried out at temperatures of about 60°C for approximately 1-4 hours. In a typical process an SO-85% conversion to polymer is obtained. • Flow diagram-	5 2	5+2	7

4	*Polymerisation in bulk* Bulk polymerisation is extensively used in the manufacture of **the** sheet and to alesser extent rod and tube.In order to reduce the shrinkage in the casting cell, and also to reduce problemsof leakage from the cell, it is normal practice to prepare a ‘prepolymer’. In atypical process monomer freed from inhibitor is heated with agitation for about8 minutes at 90°C with 0.5% benzoyl peroxide and then cooled to roomtemperature. Plasticiser, colouring agents and ultraviolet light absorbers may be incorporated at this stage if required.Acrylic sheet is prepared by pouring the syrup into a casting cell. This consistsof two plates of heat-resistant polished glass provided with a separating gasketround the edges. The gasket commonly consists of a hollow flexible tube made from a rubber, or from plasticised poly(viny1 alcohol). The cell is filled byopening up the gasket at a corner or edge and metering in the syrup, care beingtaken to completely fill the cell before closing up the gasket. The cell is heldtogether by spring-loaded clamps or spring clips so the plates will come closertogether as the reacting mixture shrinks during polymerisationThe filled cells are then led through a heating tunnel. In a typical system thetime to pass through the tunnel is about 16 hours.	7 2 2	7 2+2+3	7 7
5	Monomers of polycarbonates are bisphenol –A and phosgene Bis-phenol A may be produced by the condensation of phenol with acetone under acidic conditions  Phosgene, employed in both of the main processes, is prepared commerciallyfrom carbon monoxide and chlorine. Polymerisation Typically in such a process the bis-phenol **A** is dissolved in about ten times itsweight of pyridine and vigorously stirred at 25-35°C. Phosgene is then bubbledinto the solution and in a few minutes the pyridine hydrochloride starts toprecipitate. **As** polymer is formed the viscosity of the solution increases andeventually becomes too great for stimng. The polymer is then recovered by theaddition of a solvent such as methyl alcohol which dissolves the pyridinehydrochloride but precipitates the polymer Polytetrafluoroethylene is a linear polymer free from any significant amount of Branching	3 1+1 5	1+1+5	7

6	 PTFE is a tough, flexible, non-resilient material of moderate tensile strength butwith excellent resistance to heat, chemicals and to the passage of an electriccurrent. It remains ductile in compression at temperatures as low as 4K*(-269°C)*	2		7
	The intermolecular attraction between PTFE molecules is very small, the computed solubility parameter being 12.6 (MJ/m3)''2. The polymer in bulk doesnot thus have the high rigidity and tensile strength which is often associated withpolymers with a high softening point.			
	The carbon-fluorine bond is very stable. Further, where two fluorine atomsare attached to a single carbon atom there is a reduction in the C-F bonddistance from 1.42 A to 1.35 A. As a result bond strengths may be as high as 504 kJ/mole. Since the only other bond present is the stable C-C bond, PTFE hasa very high heat stability, even when heated above its crystalline melting point of327°C. Because of its high crystallinity and incapability of specific interaction, thereare no solvents at room temperature.	5	2+5	7
7	Melamine and formaldehyde are the monomers of MF resin Melamine (I ,3,5-triamino-2,4,6-triazine) was first prepared by Liebig in 1835. For a hundred years the material remained no more than a laboratory curiosity until Henke14 patented the production of resins by condensation with formaldehyde.			
	Today large quantities of melamine-formaldehyde resins are used in the manufacture of moulding compositions, laminates, adhesives, surface coatingsand other applications. Although in many respects superior in properties to theurea-based resins they are also significantly more expensiveReaction of melamine with neutralised formaldehyde at about 80-100°C leads tothe production of mixture of water-soluble methylolmelamines. These hydroxymethylderivatives can possess up to six methylol groups per molecule andinclude trimethylolmelamine and hexamethylolmelamine Themethylol content of the mixture will depend on the melamine formaldehyde ratioand on the reaction conditions.On further heating the methylolmelamines condense and a point is reached where hydrophobic resin separates out on cooling	2	2+5	7
	 The novolaks are prepared by reacting phenol with formaldehyde in a molar ratioof approximately 1 :0.8 under acidic conditions. Both novolaks and resols are prepared in similar equipment, shown	5		

8	diagrammaticallyin Figure. The resin kettle may be constructed from copper,nickel or stainless steel where novolaks are being manufactured. Stainless steelmay also be used for resols but where colour formation is unimportant thecheaper mild steel may be used. **In** the manufacture of novolaks, I mole of phenol is reacted with about 0.8mole of formaldehyde (added as 37% w/w formalin) in the presence of some acidas catalyst. A typical charge ratio would be: Phenol 100 parts by weight Formalin (37% w/w) 70 parts by weight Oxalic acid 1.5 parts by weight The reaction mixture is heated and allowed to reflux, under atmospheric pressure at about 100°C. At this stage valve A is open and valve **B** is closed.Because the reaction is strongly exothermic initially it may be necessary to usecooling water in the jacket at this stage. The condensation reaction will take anumber of hours, e.g. 2-4 hours, since under the acidic conditions the formationof phenol-alcohols is rather slow. When the resin separates from the aqueousphase and the resin reaches the requisite degree of condensation, as indicated by refractive index measurements, the valves are changed over (valve A **is** closed and valve B opened) and water present is distilled off.	1+1 5	2+5	7
9	Polyurethane polymers are traditionally and most commonly formed by reacting a di- or tri-isocyanate with a polyol. Since polyurethanes contain two types of monomers, which polymerise one after the other, they are classed asalternating copolymers. Both the isocyanates and polyols used to make polyurethanes contain, on average, two or more functional groups per molecule. Polyurethanes are used in the manufacture of high-resilience foam seating, rigid foam insulation panels, microcellular foam seals and gaskets, spray foam, durable elastomeric wheels and tires (such as roller coaster, escalator, shopping cart, elevator, and skateboard wheels), automotive suspension bushings, electrical potting compounds, high-performance adhesives, surface coatings and sealants, synthetic fibers (e.g., Spandex), carpet underlay, hard-plastic parts (e.g., for electronic instruments), condoms,[1] and hoses Raw materials Urea is a white crystalline compound with a melting point of 132.6°C and is highly soluble in water. It is substantially cheaper than the other intermediate(formaldehyde) used in the resin preparations. Urea is prepared commercially by the reaction of liquid carbon dioxide and ammonia in silver-lined autoclaves, at temperatures in the range 135-195°C and pressure of 70-230 atm. The reaction proceeds by way of ammonium carbamate: 2NH3 + C02 NH2*CO*NH4 NH2*CO*NH4 CO(NH2)2 + H20	2 1 4	2+1+4	7

10	FORMALDEHYDE  Urea-formaldehyde resins are usually prepared by a two-stage reaction. The firststage involves the reaction of urea and formaldehyde under neutral or mildlyalkaline conditions, leading to the production of mono and dimethylolureasThe ratio of mono to dimethylol compounds will depend on theurea-to-formaldehyde ratio and it **is** important that there should be enough formaldehyde to allow some dimethylol urea formationThe first stage of resin preparation is to dissolve urea into the 36% w/w formalin which has been adjusted to a pH of **8 with caustic soda.** The blending may be carried out without heating in a glass-lined or stainless-steel reactor for about 90 minutes. In an alternative process the blending is carried out at about 40°C for 30 minutes. In some cases the pH, which may drop during reaction, is adjusted by addition of small quantities of hexamine. The urea-formaldehyde ratios normally employed are in the range 1 : 1.3 to 1 : 1.5.Only a small amount of reaction occurs in the first stage so that the solution at the end of this process contains urea,formaldehyde, and mono- and dimethylol urea, the latterin insufficient concentration at this stage to separate out. Properties of plastic materials -. Uses of plastic materials in daily life – Automobile, electrical, industrial applications	3 4	3+4	7 7

11	 *Figure 16.6.* Tower process for mass polymerisation of styrene	7	7	7
12	Prepolymerization – Polymerization in tower-temperature at various zones of tower - extrusion - pellitization	4	4+3	7
		3		

Course :PLASTIC TECHNOLOGY

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Mark Distribution

Module	hr / Module	Marks/Module $(h_i/\sum H_i) * 123$ (±5%)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	12	34	4	4	4	12	3	21	11	37
2	11	32	2	2	1	3	3	21	6	26
3	12	34	2	2	1	3	1	7	4	12
4	8	23	1	1	3	9	3	21	7	31
Total	43	123	9	9	10	30	12	84	30	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	18	15
Understanding	105	85
Applying		
Analysing		
Evaluating		
Creating		
Total	123	100

Question Wise Analysis

Course :PLASTIC TECHNOLOGY

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.02	Remembering	1	2
I.2	M2.02	Understanding	1	2
I.3	M2.02, M2.01	Remembering	1	2
1.4	M4.03	Understanding	1	2
1.5	M3.02	Remembering	1	2
1.6	M1.02	Understanding	1	2
1.7	M1.02	Remembering	1	2
1.8	M3.04	Remembering	1	2
1.9	M1.02	Remembering	1	2
II.1	M 1.01	Remembering	3	7
II.2	M2.02	Understanding	3	7
II.3	M1.04	Understanding	3	7
II.4	M1.03	Remembering	3	7
II.5	M3.01	Understanding	3	7
II.6	M4.03	Understanding	3	7
II.7	M4.03	Remembering	3	7
II.8	M4.04	Understanding	3	7
II.9	M2.01	Remembering	3	7
III.	M1.03	Understanding	7	17
IV	M1.04	Understanding	7	17
V.	M2.01	Understanding	7	17
VI	M2.03	Understanding	7	17
VII	M3.02	Understanding	7	17
VIII	M3.04	Understanding	7	17
IX	M4.02	Understanding	7	17
X	M4.02	Understanding	7	17
XI	M4.04	Understanding	7	17

XII	M4.02	Understanding	7	17
XIII	M1.01	Understanding	7	17
XIV	M2.02	Understanding	7	17